

DATA 615 - Regression: Homework - 11

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1 Problem 11.7 (skip b) Machine Speed. (p. 473. a-g (-b), 30 pts)

- In part (d), note that the predicted (fitted) values from “regressing the squared residuals against X ” are the estimated variance of the residuals. The estimated weight is $1/(\text{estimated variance})$. If you get a negative value for the estimated variance, convert it to a positive value by taking its absolute value.

Machine speed. The number of defective items produced by a machine (Y) is known to be linearly related to the speed setting of the machine (X). The data below were collected from recent quality control records.

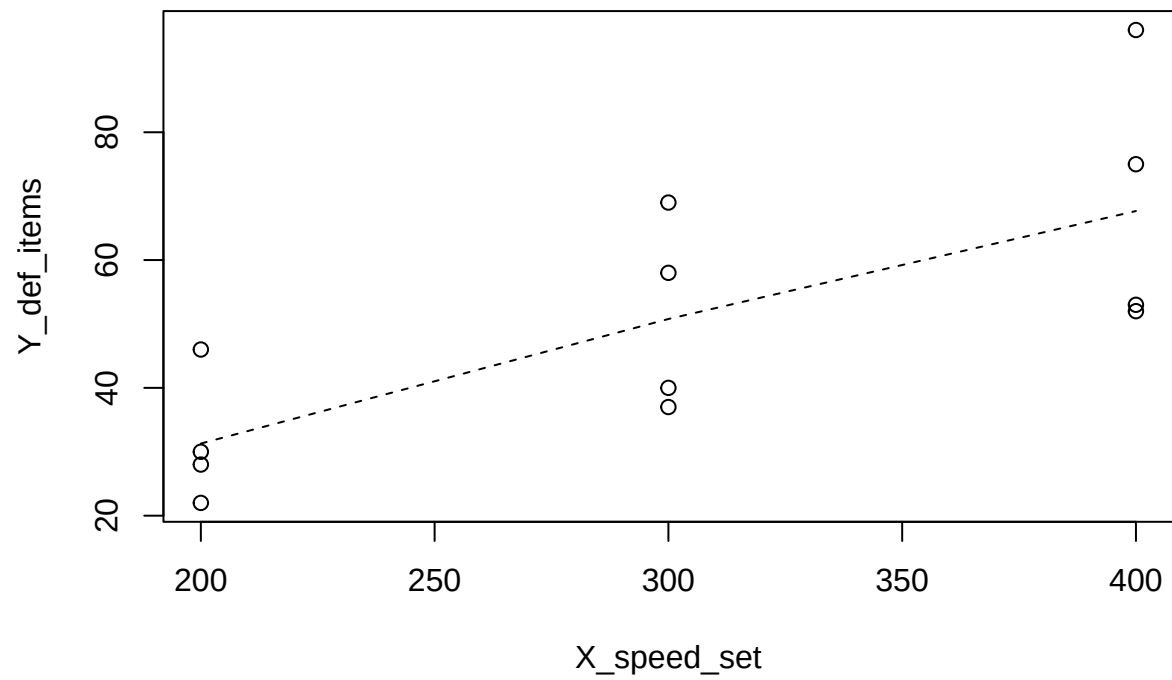
```
ms <- read.table("./data/CH11PR07.txt", header = F)
colnames(ms) <- c("Y_def_items", "X_speed_set")
glimpse(ms)
```

```
## Rows: 12
## Columns: 2
## $ Y_def_items <dbl> 28, 75, 37, 53, 22, 58, 40, 96, 46, 52, 30, 69
## $ X_speed_set <dbl> 200, 400, 300, 400, 200, 300, 300, 400, 200, 400, 200, 300
```

```
summary(ms)
```

```
##   Y_def_items    X_speed_set
## Min.   :22.00   Min.   :200
## 1st Qu.:35.25   1st Qu.:200
## Median :49.00   Median :300
## Mean   :50.50   Mean   :300
## 3rd Qu.:60.75   3rd Qu.:400
## Max.   :96.00   Max.   :400
```

```
plot(Y_def_items ~ X_speed_set, data = ms)  
lines(lowess(ms$X_speed_set, ms$Y_def_items), lty = 2)
```



a. Fit a linear regression function by ordinary least squares, obtain the residuals, and plot the residuals against X. What does the residual plot suggest?

```
ms_ols <- lm(Y_def_items ~ X_speed_set, data = ms)
ms_ols

##
## Call:
## lm(formula = Y_def_items ~ X_speed_set, data = ms)
##
## Coefficients:
## (Intercept)  X_speed_set
##      -5.7500      0.1875

ms$res1 <- ms_ols$residuals
```

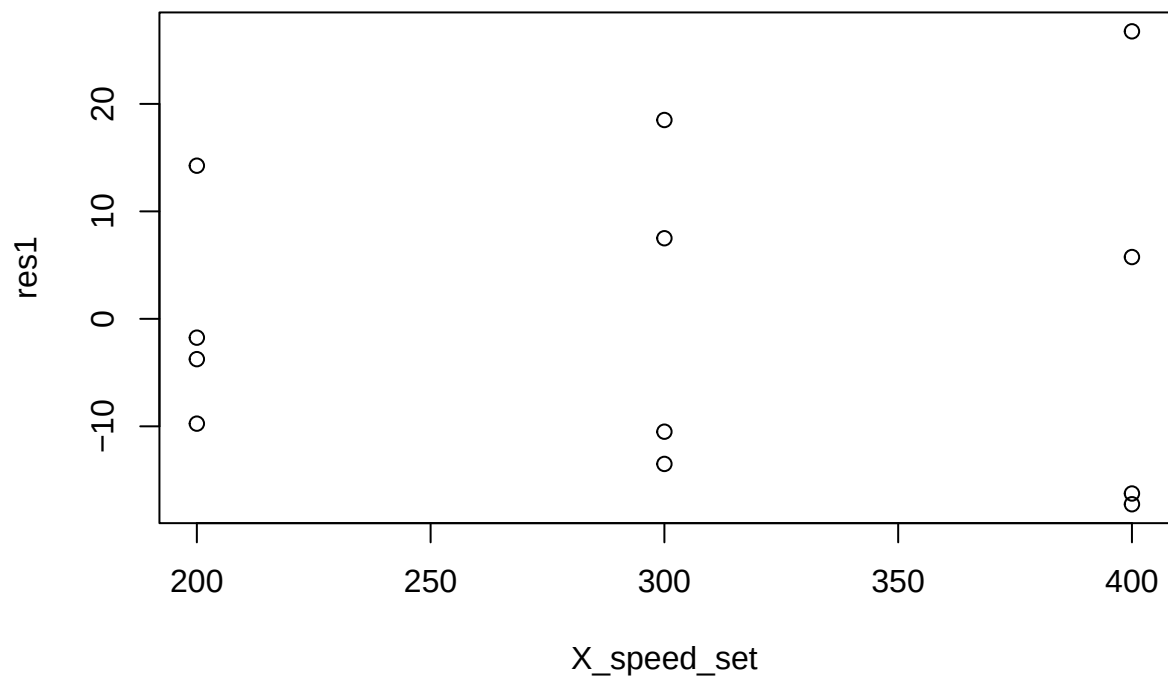
The estimated regression function by OLS is:

$$\hat{Y} = -5.7500 + 0.1875$$

The residuals:

OLS Residuals
-3.75
5.75
-13.50
-16.25
-9.75
7.50
-10.50
26.75
14.25
-17.25
-1.75
18.50

```
plot(res1 ~ X_speed_set, data = ms)
```

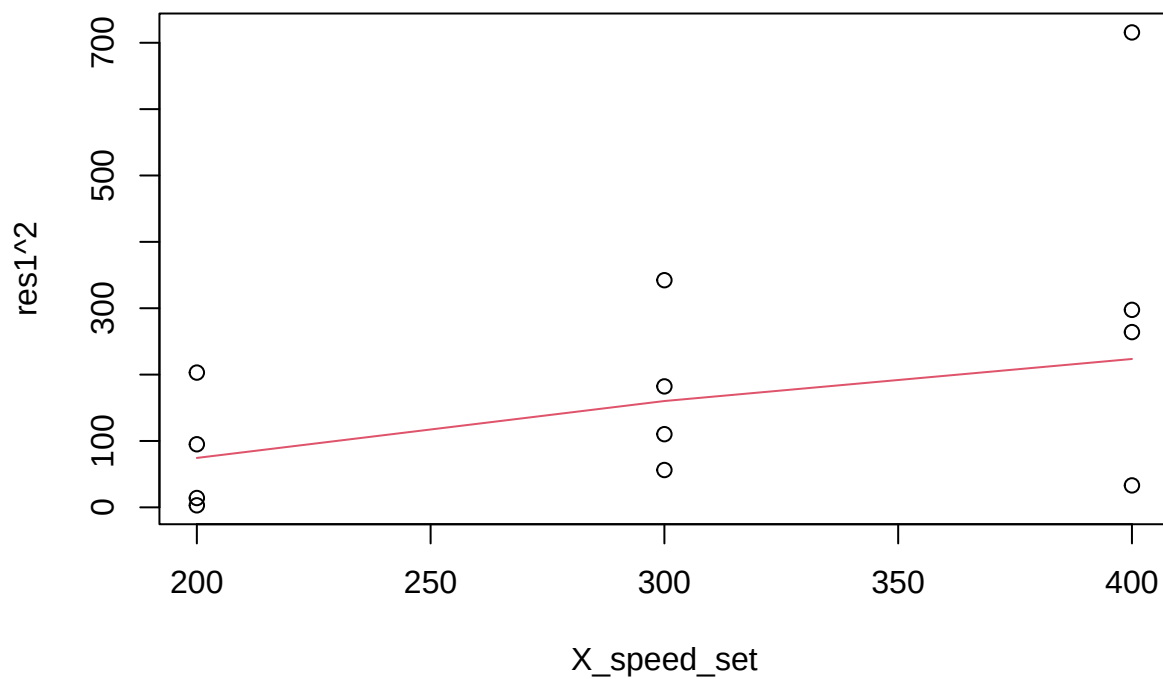


Findings: The residual plot suggests that the variance of the error terms increases as X increases and that the assumption of constant variance of the residuals is not valid.

b. skipped.

c. Plot the squared residuals against X. What does the plot suggest about the relation between the variance of the error term and X?

```
plot(res1^2 ~ X_speed_set, data = ms)
lines(lowess(ms$res1^2 ~ ms$X_speed_set), col = 2)
```



Findings: The plot indicates the variance is not constant and it increases as speed setting increases.

d. Estimate the variance function by regressing the squared residuals against X , and then calculate the estimated weight for each case using (11.16b).

```
var_ols = lm(res1^2 ~ X_speed_set, data = ms)
var_ols

##
## Call:
## lm(formula = res1^2 ~ X_speed_set, data = ms)
##
## Coefficients:
## (Intercept)  X_speed_set
##      -180.083      1.244
```

The estimated variance function is:

$$\hat{v} = -180.083 + 1.244X$$

Calculating the estimated weights:

```
ms$hat_var <- var_ols$fitted.values
ms$w1 <- 1 / (abs(ms$hat_var))
kable(ms)
```

Y_def_items	X_speed_set	res1	hat_var	w1
28	200	-3.75	68.66667	0.0145631
75	400	5.75	317.41667	0.0031504
37	300	-13.50	193.04167	0.0051802
53	400	-16.25	317.41667	0.0031504
22	200	-9.75	68.66667	0.0145631
58	300	7.50	193.04167	0.0051802
40	300	-10.50	193.04167	0.0051802
96	400	26.75	317.41667	0.0031504
46	200	14.25	68.66667	0.0145631
52	400	-17.25	317.41667	0.0031504
30	200	-1.75	68.66667	0.0145631
69	300	18.50	193.04167	0.0051802

e. Using the estimated weights, obtain the weighted least squares estimates of Beta 0 and Beta 1. Are the weighted least squares estimates similar to the ones obtained with ordinary least squares in part (a)?

```
ms_weighted_ls <- lm(Y_def_items ~ X_speed_set, weights = w1, data = ms)
ms_weighted_ls
```

```
##
## Call:
## lm(formula = Y_def_items ~ X_speed_set, data = ms, weights = w1)
##
## Coefficients:
## (Intercept)  X_speed_set
##      -6.2332      0.1891
```

The estimated weighted function is:

$$\hat{Y} = -6.2332 + 0.1891X$$

Findings: The weighted least squares estimates for β_0 and β_1 are -6.2332 and 0.1891, respectively, and they are different from the ordinary least squares estimates obtained without considering the weights.

f. Compare the estimated standard deviations of the weighted least squares estimates b_{wo} and b_{W1} in part (e) with those for the ordinary least squares estimates in part (a). What do you find?

```
summary(ms_ols)

##
## Call:
## lm(formula = Y_def_items ~ X_speed_set, data = ms)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.250 -11.250  -2.750   9.188  26.750
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -5.75000    16.73052  -0.344  0.73820
## X_speed_set  0.18750     0.05381   3.484  0.00588 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.22 on 10 degrees of freedom
## Multiple R-squared:  0.5484, Adjusted R-squared:  0.5032
## F-statistic: 12.14 on 1 and 10 DF,  p-value: 0.005878
```

```
summary(ms_weighted_ls)

##
## Call:
## lm(formula = Y_def_items ~ X_speed_set, data = ms, weights = w1)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -1.1572 -0.9338 -0.3124   0.7377   1.7391
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6.23322    13.16843  -0.473  0.64613
## X_speed_set  0.18911     0.05056   3.740  0.00385 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.109 on 10 degrees of freedom
## Multiple R-squared:  0.5831, Adjusted R-squared:  0.5414
## F-statistic: 13.99 on 1 and 10 DF,  p-value: 0.003846
```

Findings: The standard deviation for the ordinary least squares for coefficients β_0 and β_1 are: 16.73052 and 0.05381, respectively. While the standard deviation for the **weighted** least squares for coefficients β_0 and β_1 are: 13.16843 and 0.05056, respectively. It's noticeable that the standard deviation for both estimated coefficients with weighted least squares are smaller and that is because the weighted least squares estimates are unbiased and more consist compared with the results from the OLS regression.

g. Iterate the steps in parts (d) and (e) one more time. Is there a substantial change in the estimated regression coefficients? If so, what should you do?

```
ms$resi2 <- ms_weighted_ls$residuals
ms$hat_var2 <- lm(resi2^2 ~ X_speed_set, data = ms)$fitted.values
ms$w2 <- 1 / abs(ms$hat_var2)
kable(ms)
```

Y_def_items	X_speed_set	res1	hat_var	w1	resi2	hat_var2	w2
28	200	-3.75	68.66667	0.0145631	-3.588927	68.60343	0.0145765
75	400	5.75	317.41667	0.0031504	5.588927	317.51450	0.0031495
37	300	-13.50	193.04167	0.0051802	-13.500000	193.05896	0.0051798
53	400	-16.25	317.41667	0.0031504	-16.411073	317.51450	0.0031495
22	200	-9.75	68.66667	0.0145631	-9.588927	68.60343	0.0145765
58	300	7.50	193.04167	0.0051802	7.500000	193.05896	0.0051798
40	300	-10.50	193.04167	0.0051802	-10.500000	193.05896	0.0051798
96	400	26.75	317.41667	0.0031504	26.588927	317.51450	0.0031495
46	200	14.25	68.66667	0.0145631	14.411073	68.60343	0.0145765
52	400	-17.25	317.41667	0.0031504	-17.411073	317.51450	0.0031495
30	200	-1.75	68.66667	0.0145631	-1.588927	68.60343	0.0145765
69	300	18.50	193.04167	0.0051802	18.500000	193.05896	0.0051798

```
ms_weighted_ls2 <- lm(Y_def_items ~ X_speed_set, weights = w2, data = ms)
ms_weighted_ls2
```

```
##
## Call:
## lm(formula = Y_def_items ~ X_speed_set, data = ms, weights = w2)
##
## Coefficients:
## (Intercept)  X_speed_set
##      -6.2335      0.1891
```

The estimated second interaction weighted function is:

$$\hat{Y} = -6.2335 + 0.1891X$$

Findings: With the second iteration, the weighted least squares estimates for β_0 and β_1 are -6.2335 and 0.1891, respectively. The estimates didn't change much from the first interaction and that suggests that no more iterations is needed.

This is the end of Homework 11.